Kappa Foundation

Initial decision record

USD | Case date 31 December 2022 | Initial submission 22 October 2026

Investment objective and policy

Meet the fixed education commitments first and protect long-term purchasing power. Invest surplus capital within the fund's capacity for risk. Seek **USD 8m in December 2022 purchasing power at end-2030**; this is a planning goal. Portfolio changes require evidence about obligations, resources and assets, rather than short-term market moves.

Initial capital allocation



Beta 18.46% Delta 28.58% USMV 52.89% Costs 0.08%

Holding / use	Units / basis	Initial USD
Beta | 2024	2,400	2,214,672.00
Delta | 2027	3,478.26087	3,429,808.70
USMV	100% of sleeve	6,346,350.71
Trading costs	5 / 10bp	9,168.59
Total		12,000,000.00

Source: case Exhibit 1; Kappa model; USD, fees separate; % rounded.

Beta principal pays **$2.4m on 31 Dec 2024**. Delta principal and its final coupon pay **$3.6m on 31 Dec 2027**. Four earlier Delta coupons total $486,957 and remain cash at 0%; payment coverage needs no equity sale or reinvestment return.

Growth portfolio decision

Recommend 100% USMV, the US minimum-volatility ETF. Use **100% VTI**, a broad-US-market ETF, as the frozen investable benchmark and credible rejected alternative.

Annual measure	USMV	VTI	Max utility A=4
Expected return[1]	7.14%	8.45%	7.19%
Volatility[2]	12.84%	16.31%	12.70%
Utility[3] **| A=4**	3.84%	3.13%	3.96%

Source: Yahoo Finance, Kenneth French; Kappa model; 2015-22, annualised.

Full-universe maximum utility at A=4: MSFT 2.92% | JPM 3.52% | JNJ 7.46% | PG 11.62%
XOM 2.96% | NEE 7.45% | USMV 46.18% | VTI 17.89% | AAPL/CAT 0.00%

This is the highest-scoring long-only mix of the 10 candidates for these inputs: 96 months, CAPM expected returns, sample covariance, weights >=0 and sum=100%, no extra caps. At A=4 it estimates 7.19% return, 12.70% volatility and 3.96% utility, 0.12 percentage point above USMV. But the 96-to-60-month check shifts 34.8% of weights one-way (about $2.21m on this sleeve). We choose USMV for simpler policy implementation, not because it is the model maximum.

Neither portfolio guarantees principal or the 2030 goal. The optimizer's weights are estimates, not an instruction to trade without review.

Long-only; no borrowing, shorting or derivatives. Fractional holdings allowed. Fee sensitivity: 0bp bonds / 5bp ETFs / 10bp stocks.

Mandate facts

Initial capital	$12.000m
Currency	USD
Goal date	31 Dec 2030
Real goal	$8.000m
Growth sleeve	100% USMV
Benchmark	100% VTI

Source: original case and proposed IPS.

Core assumptions

CPI base	2.50%
HEPI stress	3.00%
Risk-free (plan)	4.00%
Market premium	4.50%
Forecast alpha	0.00%
Risk aversion A	4 (provisional)
Coupon cash rate	0.00%
Bond / equity fee	5 / 10bp

Source: proposed IPS; annual scenarios.

Candidate universe

Technology	AAPL, MSFT
Financials	JPM
Health care	JNJ
Staples	PG
Energy	XOM
Industrials	CAT
Utilities	NEE
Equity ETFs	USMV, VTI

Source: eligibility notes; 2022 status.

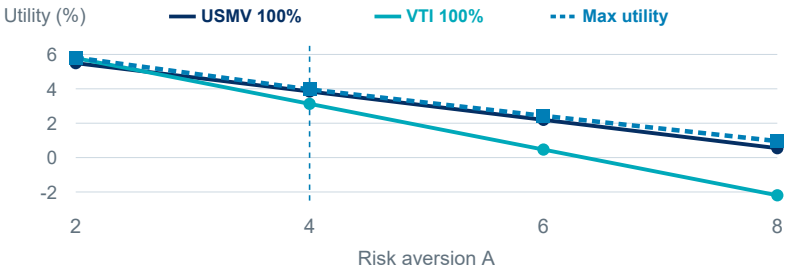
Eight stocks, seven sectors and two distinct ETF mandates. Candidates need not all be held. No additional security-weight caps are imposed.

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Policy, risk and implementation

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Risk-preference comparison



Source: Kappa analysis; Yahoo/French inputs; annual utility scores.

USMV and VTI stay at 100%. The dashed line re-optimises all 10 candidates at each A and shows the highest estimated score for that risk preference. At A=4 it scores 3.96%; page 1 lists its holdings. Of the fixed ETFs, VTI leads at A=2; USMV leads at A=4, 6 and 8 (crossover A=2.58). A=4 is provisional.

Real-goal sensitivity

2030 goal	Base	Stress
Inflation	2.50%	3.00%
Nominal USD	$9.747m	$10.134m
CAGR[4] hurdle	4.84%	5.37%

Source: funding model; 2022-30.

CPI defines the primary goal. HEPI is a separate education-cost stress. Both hurdles retain four early coupons at 0% cash interest.

Risk oversight | growth sleeve

Use flow-adjusted growth-sleeve unit values, including cash; retain the prior peak after asset changes. Report payment coverage and whole-fund stress separately. Historical drawdown[5] is measured at month-end.

Signal[5]	Required action	Deadline
10%	Validate data, funding and stress	Within 2 business days
15%	Investment Committee decision	Within 5 business days
20%	Notify trustees; pause risk increases	Same / next business day
Payment gap	Escalate; restore payment coverage	Immediate

Source: proposed IPS, 21 Oct 2026; growth-sleeve review triggers, not guaranteed loss limits.

No automatic sale. At 15%, record hold, authorised rebalance or policy proposal. Discretionary crisis deferral requires written authority, expiry and review; due-payment coverage cannot fall.

Annual goal protection and policy response

At year-end, after reserving remaining bond/cash obligations, test the **$9.747m** CPI goal (**$10.134m** at HEPI stress) after a **30% growth-sleeve loss**. If covered, trustees may approve locking only the minimum needed in an eligible, priced asset due by 2030; leave surplus invested. This is prospective, not automatic. If not covered, report the gap; do not raise risk to catch up.

Frozen test and future rebalancing

Freeze **USMV 100%** and **VTI 100%** at 31 Dec 2022. Compare monthly returns in Jan 2023-Dec 2025; reset to start weights each December without re-estimating. One ETF creates no internal trade; 10bp applies only to trades. Later returns test, not set, 2022 weights. Future changes require a dated IPS review.

Estimate 96 months (Jan 2015-Dec 2022); check 60 months. The final memo compares sample and single-index risk. FF3 explains history; CAPM sets forecast alpha to zero. Exclude 2023-25 from 2022 weights.

Read and reproduce

1. Read **data/DATA_RECORD.md** and its retrieval manifest. **2.** Install requirements-analysis.txt; run **python analysis/run_initial.py** offline. **3.** Check **sample_utility_robustness.json**, **max_utility_A4_weights.csv** and **max_utility_preference.csv** in outputs/. Retrieval: **analysis/retrieve_data.py --phase train**, in a clean copy.

Data retrieved 15 Oct 2026: Yahoo adjusted-close and French factors. Revised histories and survivor selection limit this retrospective case. Fictional bonds use contract cash flows. RF=4% is a planning scenario; historical excess returns use observed monthly French RF.

Mini-Glossary | Definitions used in this record

1 CAPM return	Annual arithmetic forecast: risk-free rate + beta x market premium; forecast alpha=0.
2 Volatility	Monthly return standard deviation x sqrt(12); an annual risk estimate, not a loss cap.
3 Utility (A)	E[R] - 0.5 x A x annual variance; a higher A applies a larger penalty to risk.
4 CAGR	The constant annual compound rate over a period; the goal hurdle includes retained coupon cash.
5 Drawdown	Decline in the flow-adjusted growth-sleeve unit value from its previous peak.